

Louisville Labor Market Monitor Project Proposal

Developer: Hannah Sadikovic

Target Audience: Louisville, KY residents and visitors

Technology Stack: Python, PostgreSQL, Power BI, BLS API

Background

This project proposes the development of an interactive labor market monitor that shows trends and comparisons of labor market indicators and industry employment in Kentucky and the Louisville metropolitan area. The dashboard utilizes economic data from the U.S. Bureau of Labor Statistics (BLS) Public API to monitor monthly labor market trends and employment patterns.

Business Problem

Economic anxiety is currently very high and many fear potential economic instability and labor market uncertainty. This project attempts to inform residents of Louisville and Kentucky about current labor market conditions through an interactive dashboard. Individuals considering relocation, home purchases, career changes, or industry transitions can use these labor market indicators to make more informed decisions.

Objectives

- Extract and integrate monthly labor market data from the BLS API using Python
- Transform and clean JSON data into a structured relational format
- Store processed data in PostgreSQL
- Create an interactive labor market dashboard using Power BI
- Provide users with continuously updateable labor market indicators and employment trends

Technical Approach

Kentucky Labor Market Dashboard

- Unemployment Rate
- Employment Level
- Labor Force

Louisville Industry Employment Dashboard

Industry employment data for five major Louisville metropolitan industries:

- Manufacturing
- Private Education and Health Services
- Transportation and Warehousing

- Retail Trade
- Government

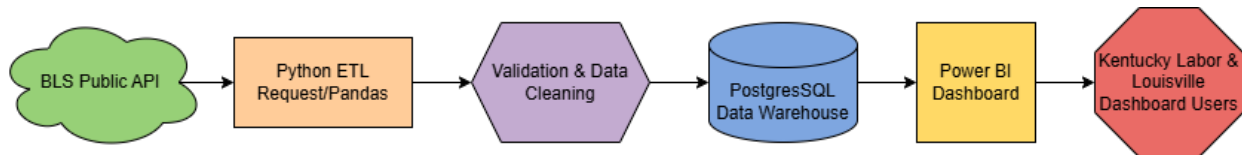
Interactive User Interface

- Year filtering
- Industry filtering
- Interactive Power BI visualizations

Data Flow

1. API Integration: Fetch labor market data from the BLS API
2. Data Processing: Transform raw JSON data into structured tabular formats
3. Data Storage: Load validated data into PostgreSQL
4. Analytics Layer: Connect Power BI directly to PostgreSQL
5. User Interaction: Allow filtering and exploration of labor market trends

Data Flow Diagram



Visualization Design

Page 1: Kentucky Labor Market Overview

- Highest Unemployment Rate
- Lowest Unemployment Rate
- Average Annual Unemployment Rate
- Unemployment Rate Trend
- Year Filter

Page 2: Louisville Industry Employment Analysis

- Employment Trends by Industry
- Louisville Employment by Industry
- Industry Filter

Data Source and API

Primary Data Source: BLS Public API V2

- Authentication key required

- Series ID formats for state and metropolitan labor market data

Kentucky Labor Market Metrics

- Unemployment Rate
- Employment Level
- Labor Force

Metric	Series ID
Unemployment Rate	LASST2100000000000003
Employment Level	LASST2100000000000004
Labor Force	LASST2100000000000005

Louisville Metropolitan Industry Employment

Area: Louisville/Jefferson County, KY-IN Metropolitan Area (31140)

Area	Area Code
Louisville/Jefferson County, KY-IN Metropolitan Area	31140
Kentucky	21

Industries

- Manufacturing
- Transportation and Warehousing
- Private Education and Health Services
- Retail Trade
- Government

Industry	Series ID
Manufacturing	SMU213114030000000001
Transportation and Warehousing	SMU213114043000000001
Private Education and Health Services	SMU213114065000000001
Retail Trade	SMU213114042000000001
Government	SMU213114090000000001

Data Storage

Primary Storage: PostgreSQL Database

Analytics Layer: Power BI

Python Libraries:

- requests

- pandas
- sqlalchemy
- psycopg2

Data Cleaning Techniques

Use Python to:

- Handle missing values
- Convert data types
- Validate observation values
- Remove duplicate records
- Standardize date fields
- Perform data quality checks

Timeline

Week	Tasks
Week 1	Research topic, select API, draft proposal
Week 2	Design relational database schema and build ETL pipeline
Week 3	Complete PostgreSQL storage layer and validation processes
Week 4	Build Louisville Labor Market dashboard with visualizations and filters
Week 5	Final testing, presentation, and submission

Expected Outcomes

- Documented and working ETL pipeline using Python, PostgreSQL, and the BLS API
- Validated labor market data loaded into a relational database
- Interactive Power BI dashboard for Kentucky labor market indicators and Louisville industry employment trends
- Executive presentation of findings
- Organized GitHub repository with project documentation

Future Enhancements

Potential future enhancements include:

- Demographic analysis by age and gender
- Additional Louisville labor market series
- Wage and earnings analysis
- Automated monthly refresh scheduling
- Expanded industry coverage